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Thermal performance in latticework ducts with various endwall shapes for aero-craft turbine cooling

Lei Luo [a], Wei Du [a,*], Jinlong Liu [b], Peipei Sun [b], Songtao Wang [a], Bengt Sunden [c,*]

[a] School of Energy Science and Engineering, Harbin Institute of Technology, Harbin, China
[b] Aero Engine Academy of China, Beijing, China
[c] Department of Energy Sciences, Lund University, Lund, Sweden

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ABSTRACT

The latticework duct plays an important role in turbine blade cooling for aircraft engines. The present numerical research provided information about the thermal performance in latticework ducts with different endwall shapes. Each latticework duct analyzed in this study had a crossing angle of 90° and 11 sub channels on both the two sides of the endwall. A numerical investigation was conducted at five Reynolds numbers (11,000 to 55,000) and six endwall shapes. The local *Nu* number and flow structures were analyzed in detail. For a rectangular latticework duct, a jet region generated considerable Nusselt numbers, and the helical flow leaded to a uniform Nu number. Moreover, the turning and jet effects caused a high friction factor. The results also showed that an appropriate endwall shape could increase the thermal performance in a latticework duct. The different endwall shapes mainly altered the flow structure nearby the sidewall of the latticework, especially for the spiral flow. Moreover, the endwall shape also influenced the heat transfer and flow behavior. This phenomenon occurred because the different endwall shapes increased the turbulence of the cooling air to different extents.
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1. Introduction

The thermal efficiency of jet engines can be improved by raising the turbine inlet temperature. However, the inlet temperature for advanced gas turbines has already passed 1700 °C [1], which greatly surpasses the tolerance temperature of metal and ceramic matrix composites. Consequently, to protect the turbine blade, many cooling techniques, such as the use of jets [2], U-bend [3], pin fins [4], dimples [5], film cooling [6] and double wall [7] have been developed since the 1970s, and a considerable number of high-quality reviews have been published in regard to these cooling structures in gas turbines [1,8,9]. Nevertheless, these cooling structures are utilized primarily by GE (General Electric Company), P&W (Pratt & Whitney Group) and RR (Rolls-Royce Plc), whereas the latticework duct, which is also characterized as matrix cooling or bounded vertical cooling, is usually applied in Soviet turbine blades. The latticework duct includes many sub channels within the blade, as shown in Fig. 1 [10], where a sub channel comprises two repeated- parallel- rib layers. These repeated parallel ribs are responsible for high heat transfer capability and structural strength. In addition, the heat transfer coefficient within the turbine blade is uniformly distributed, corresponding to a low thermal stress.

A review of the available literature about latticework ducts reveals that a crossing angle of 90° between the two layers of ribs was found to induce the highest Nusselt (*Nu*) number [11]. Therefore, the crossing angle was fixed at 90° for most follow-up studies. However, since 2000, only a few studies have been published about latticework ducts. Gillespie et al. [12] put a latticework duct in a wedge duct (flow area is gradually decreased in the flow direction) and imaged the Nu number distribution by employing the transient liquid crystal technique. Their results proved that the selected latticework duct greatly enhanced the heat transfer ability. Similarly, Bunker [13] experimentally measured the heat transfer and showed that the Nu number distribution is uniform in the sub channels. Bunker [13] also used infrared thermography methods to test the Nu number distribution; the results obtained by the liquid crystal technique were similar to those obtained by infrared thermography. Acharya and Bunker [14] further explored the thermal performance in a latticework duct operated in rotating conditions. It was found that the rotation number only has a minor impact

* Corresponding authors.
E-mail addresses: hitdw9211@outlook.com (W. Du), bengt.sunden@energy.lth.se (B. Sunden).

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Nomenclature

H_i	Amplitude of the endwall shape ..... mm	T_w	Wall temperature ..... K
L	Latticework duct length ..... mm	W_i	Rib spacing ..... mm
q	Wall heat flux ..... W/m^2	Greek Symbols	
t	Rib width ..... mm	β	Rib crossing angle ..... °
T_b	Bulk temperature ..... K	υ	Local kinematic viscosity of the fluid
T_{in}	Inlet temperature ..... K		

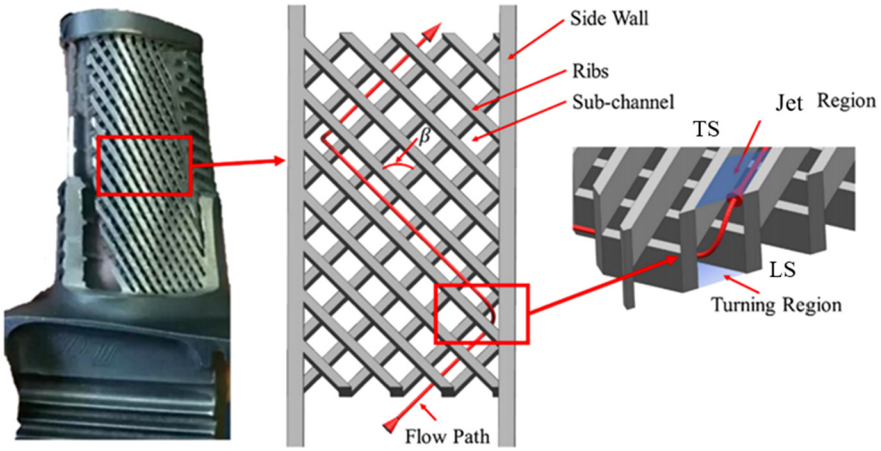


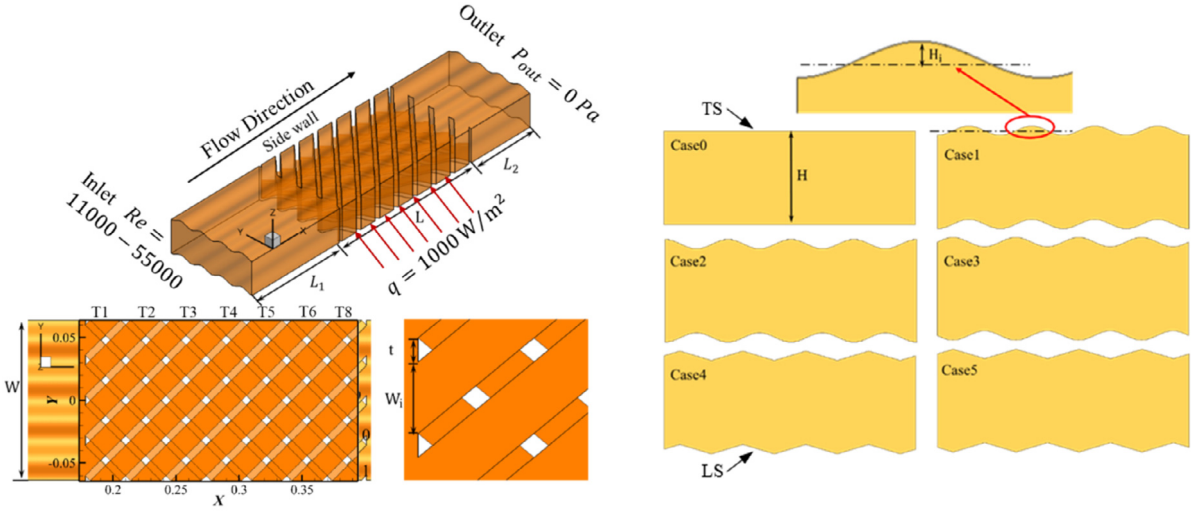
Fig. 1. Cutaway model of the AL-31 turbine blade [10].

on the thermal performance when the Reynolds number (Based on the inlet hydraulic diameter) was greater than 20,000. Saha et al. [15] installed a latticework in a wedge channel. To improve the measurement resolution of the Nu number, a narrow-band liquid crystal was applied. The confirmed that a latticework duct obtains a uniform Nu number; These findings agree well with those reported by Bunker [13]. Furthermore, Saha et al. [15] demonstrated that the ending of a sub channel contributes to the high-pressure gradient, which corresponds to high energy loss. Oh et al. [16] utilized the naphthalene sublimation method to study the thermal performance. Their findings illustrated that the Nu number distribution is dominated by turning, jet and swirling flows. Under rotating conditions, Nu number is sensitive to the rotation number on the leading side (LS), while the Nu number is insensitive on the trailing side (TS). Deng et al. [17] changed the geometry near the sidewall to account for the effects of turning and jet flows therein to enhance the Nu number. Their results suggested that the heat transfer was improved after incorporating a slot near the sidewall. Evidently, previous studies display a lot of information about heat transfer characteristics in latticework ducts; In contrast, less information is available about the flow characteristics due to the complex cooling structure. Fortunately, an increasing amount of flow structure investigations is being reported with the development and application of experimental techniques and numerical approaches. The magnetic resonance imaging technique was used to reveal the flow characteristics by Tsuru et al. [18]. Their detailed experimental results illustrated the bypass, turning and helical flow. Moreover, Bu et al. [19] numerically studied the flow structure and heat transfer distributions of a latticework duct that combined slots and pin fins; Their results displayed large vortices in each sub channel which contribute to the strong swirling motion in a latticework duct.

The above scrutinizing of the literature illustrates that previous researches focused only on exploring the flow fields and heat transfer in rectangular latticework ducts. Fig. 1 shows that a latticework duct is formed by many ribs. These ribs are placed on the endwall surface on both the LS and the TS. According to previous studies, the ribs crossing angle, rib thickness and rib number have an impact on heat transfer. However, only limited research has been conducted on the thermal behaviors of such ducts with different endwall shapes. An appropriate endwall shape can change the secondary flow and significantly enhance heat transfer. Thus, it is important to explore whether different endwall shapes can change the thermal performance in a latticework duct. Accordingly, this study numerically investigates the thermal performance of latticework ducts with different endwall shapes.

2. Geometry

To study the various impacts of different endwall shapes (The top wall or bottom wall of the latticework duct), a simplified latticework duct is utilized, as shown in Fig. 2. The width of the latticework duct (W) is 24 mm, and the length (L) is 224 mm. The duct is segmented into many sub channels by ribs on both the two sides. The rib width (t) is 8 mm, and the ribs spacing (W_i) is 24 mm. The crossing angle (β) is 90°. 11 sub channels are located on the channel. For the sake of convenience, the sub channels are numbered from T1 to T11 and L1 to L11 on the two sides, respectively. In addition, two extension parts ($L_1 = 100$ mm, $L_2 = 80$ mm) are placed before the inlet and outlet, respectively. To show the influence of the endwall shape on heat transfer, six different endwall are designed, as shown in Fig. 2(b). The inlet cross-section is rectangular with a height (H) of 48 mm at Case 0. The endwall shape for Case 1 through Case 3 is described by a sine function and cosine function with an amplitude (H_i) of 2.4 mm on both the trailing side and the leading side. For Cases 4 and 5, the endwall shape is determined by a broken line. These different endwall shapes alter the shape of the flow passage area. In particular, different endwall shapes clearly modify the turning and jet regions. Table 1 shows the characteristics of the different cases.



(a) Global view of the latticework (b) Different shapes of the latticework

Fig. 2. Geometry model.

Table 1
Geometry parameters.

	Case 0	Case 1	Case 2	Case 3	Case 4	Case 5
Shape	Straight line	Sine and cosine ($H_i = 24$)			Broken line($H_i = 24$)	
Trailing side	constant	expansion	shrink	expansion	expansion	expansion
Leading side	constant	shrink	shrink	expansion	shrink	expansion

3. Computational details

3.1. Parameter definitions

The inlet hydraulic diameter is defined as follows:

$$D_h = \frac{4 \times W \times H}{2(W + H)} \tag{1}$$

where W, H are the width and height of the duct, respectively.

The Reynolds number is written as:

$$Re = \frac{\rho U_{in} D_h}{\mu} \tag{2}$$

where ρ, μ, U_{in} are the density, dynamic viscosity, and inlet average velocity at the inlet.

The local Nu number in the latticework duct is obtained as follows:

$$Nu = \frac{q}{T_w - T_b} \cdot \frac{D_h}{\lambda} \tag{3}$$

where q, T_w, T_b are the wall heat flux, wall temperature and local bulk temperature, respectively.

The Nu number in the latticework duct is compared with the follow equation [20]:

$$Nu_0 = 0.023 Re^{0.8} Pr^{0.4} \tag{4}$$

The friction factor is calculated by:

$$f = \frac{(P_{in} - P_{out}) D_h}{0.5 \rho U_{in}^2 L} \tag{5}$$

The friction factor is non-dimensionalized by using [21]:

$$f_0 = 0.316 Re^{-0.25} \tag{6}$$

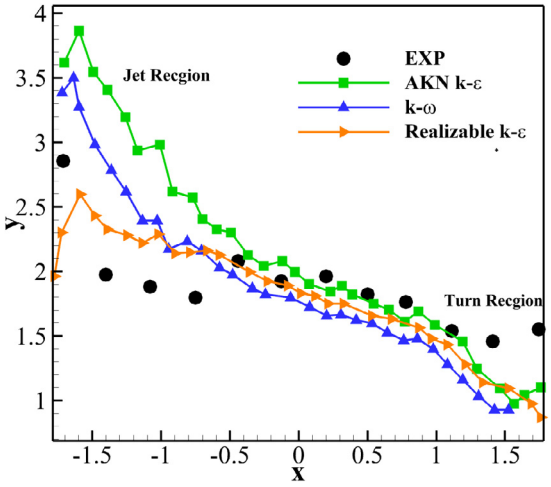
3.2. Numerical validation

Numerical validation is crucial for any numerical investigation to ensure the accuracy of the calculation. In this study, five different turbulence models, available in the STAR CCM+, are considered for comparison with the experimental results [16]. A polyhedral mesh discretizes the computational domain. A Reynolds number $Re = 44{,}000$ is selected as the working condition. Fig. 3 shows the laterally averaged Nu number distributions along the flow direction obtained by the numerical method and experimental method.

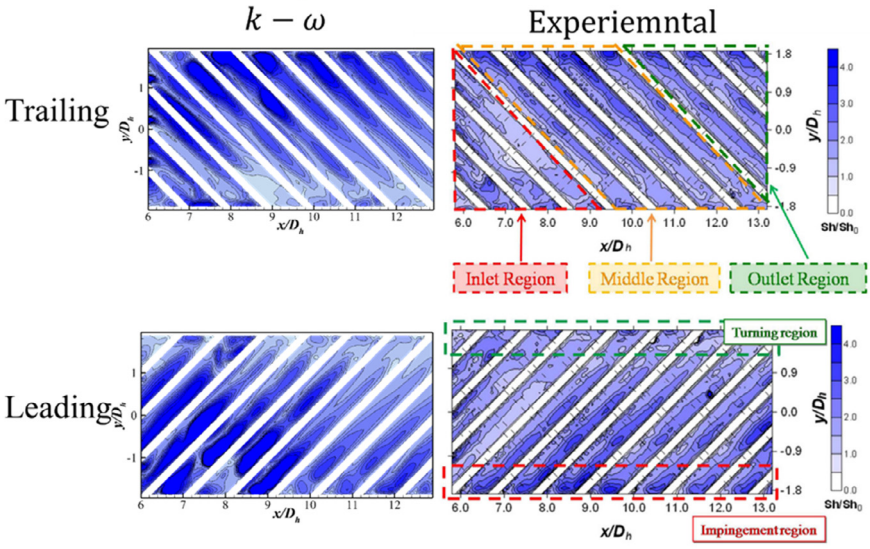
Beyond the jet region, the Nu number exhibits a downward trend in the experimental results. All of the turbulence models capture this decreasing trend. Nearby the turning region, the Nu number exhibited an increase. Only the $k-\omega$ and AKN turbulence models can capture this characteristic. However, the $k-\omega$ turbulence model has better performance near the jet region. The maximum deviation is around 40%, which is higher than the experimental uncertainty (8.5%). However, the trend of the Nusselt number distributions calculated by the k-ω turbulence model fits well with the experimental results. The Fig. 3(b) shows the local Nu number distributions on endwall surface. This also proves that the $k-\omega$ turbulence model is a suitable model for this study. Table 2 shows a comparison of the friction factor with different model. It is found that the $k-\omega$ turbulence model provides acceptable accuracy for friction factor. The deviation between the experimental result and the $k-\omega$ turbulence model is 4.56%. Thus, the $k-\omega$ turbulence model is chosen in this study according to the heat transfer characteristics and friction factor.

3.3. Boundary conditions

For the present research, the Re is from 11,000 to 55,000, corresponding to different velocity. The uniform velocity and temperature ($T_{in} = 20\,^{\circ}C$) are used for the inlet. The reference pressure in this study is atmospheric pressure ($P_{ref} = 101.325$ kPa), therefore, the pressure for the outlet is $P_{out} = 0$ Pa. The wall heat flux



(a) A comparison of the laterally averaged Nu number distributions along the flow direction between the experimental results and numerical results



(b) Local Nu number distributions

Fig. 3. Numerical validation. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 2
Friction factor with different turbulence model.

Method	Experimental result	AKN	k − ω	Realizable
f/f_0	4.6	4.28	4.39	4.21

for the TS, LS and sidewall is q = 1000 W/m^2. To diminish numerical dissipation, a 2nd order is selected to solve the turbulent flow [22]. The convergence criterion for the flow field is that the residual should be smaller than 10^{-4} and for the energy equation the residual should be smaller than 10^{-6}.

3.4. Mesh information

In this study, a polyhedral mesh is used. A polyhedral mesh requires a smaller mesh number and provides greater accuracy than a tetrahedral mesh [23]. Fig. 4(a) shows the mesh distribution and quality in this study. To capture the boundary flow, a 20-layer prismatic grid is used near the wall. The height of the first layer is 0.008 mm, corresponding to $Y^+ = 1$. The Y^+ distribution at the endwall is displayed in Fig. 4(b). Furthermore, the growth ratio for the prismatic grid is 1.15. To decide the proper mesh number, a mesh independence study is also performed, as shown in Table 3. It was found that 7.0 million nodes are enough to ensure precision both on the endwall and rib surface. Therefore, 7.0 million nodes are used to discretize the computational domain.

4. Results and discussion

4.1. Overall performance

Fig. 5 shows the overall thermal performance for the lattice-work ducts with different Reynolds numbers and endwall shapes. Fig. 5(a) exhibits the area-averaged Nu number on the trailing side. It is found that the value of the Nu number is strongly associated with the endwall shape. At *Re* = 11,000, Case 0 shows the lowest Nu number and Case 3 shows the largest values. However, Case 1 results in the lowest Nu number, Case 3 results in the highest when *Re* is greater than 11,000. Compared with the Nu number in Case 0, Nu number in Case 1 decreases by 6.4%, whereas the Nusselt numbers in Case 2, Case 3, Case 4, and Case 5 increase by 2.0%, 6.8%, 1.8% and 2.5%, respectively. This suggests that the large flow area near the sidewall has high Nusselt number. Furthermore, the value of *Nu* number among the different cases gradually be-

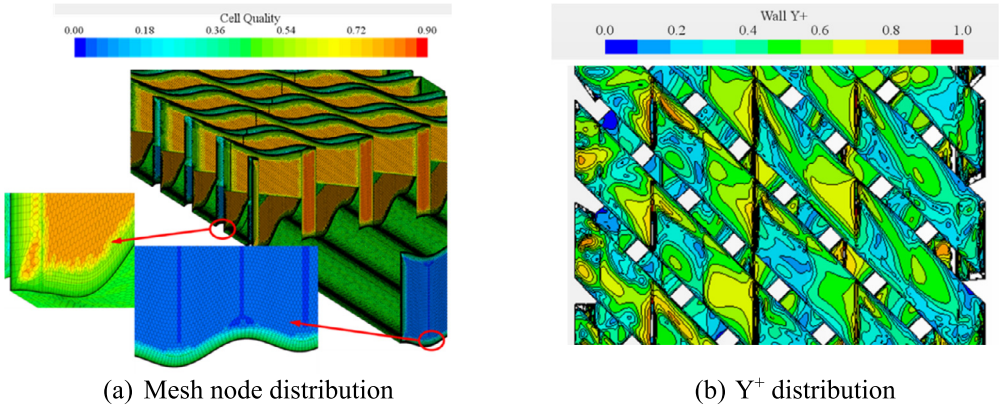


Fig. 4. Mesh information.

Table 3
Mesh independence.

Mesh number	6.0 million	7.0 million	8.0 million	9.0 million
Nu on the pressure side	180.02	186.22	185.21	186.92

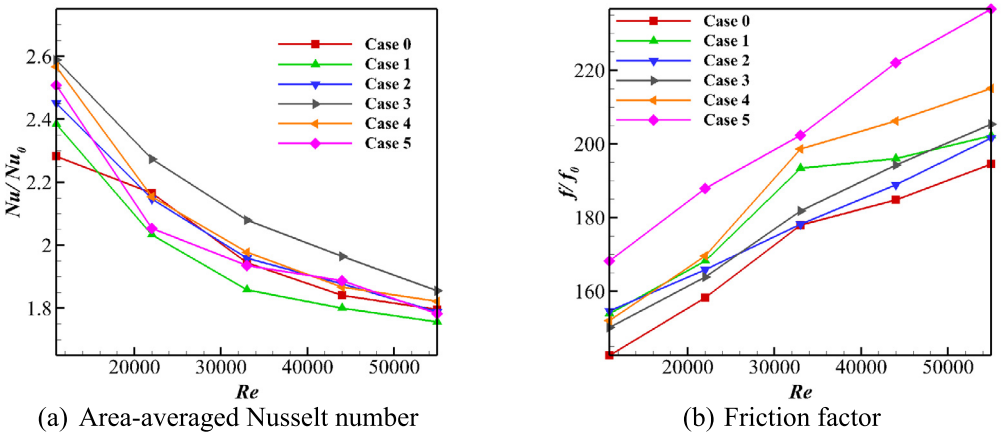


Fig. 5. Overall performance of the latticework with different Reynolds numbers and endwall shapes.

comes similar as *Re* increases. Besides, the *Nu* number decreases as *Re* increases for all cases. This characteristic is consistent with a passive heat transfer enhancement device. Regarding the friction factor, Case 0 owns the lowest value, whereas Case 5 displays the highest. This finding indicates that modifying the endwall shape of the latticework duct is responsible for the high friction factor. The friction factors in Case 1, Case 2, Case 3, Case 4 and Case 5 increase by 8.7%, 10.2%, 5.6%, 11.6% and 18.7%, respectively, compared with the friction factor in Case 0. Thus, the smooth endwall shape can result in low friction factor. Moreover, a high *Re* number can promote an upward trend in the friction factor.

4.2. Heat transfer distributions

Fig. 6 displays the local *Nu* number at $Re = 33{,}000$. Modifying endwall shape clearly alters the Nu number distribution. For Case 0, the inlet part exhibits lower Nusselt numbers than the other parts of the duct due to the absence of the turning and jet effects. Nevertheless, in the inlet part, high Nu numbers, which are attributed to the horseshoe vortex, are found at the entrance of the latticework duct. In the middle part, the Nusselt numbers sharply increase compared with those in the inlet part. In the middle part, the Nu number is decreased follow the Y direction. In the outlet part, the Nu numbers slightly decrease. For Case 1, the horseshoe vortex weakens in the inlet part. Thus, Nu number at the inlet part for Case 1 is less than that in Case 0. For the middle part, the Nu number near the jet region becomes greater than that in Case 0. However, these high Nu numbers decrease sharply along the flow direction because of the high flow resistance encountered therein. Because of the disturbance of the endwall surface, another region of relatively high Nusselt numbers is observed between the peak and trough of the endwall in Case 1. The Nu between the Case 2 and Case 1 is similar; however, the region of relatively high Nusselt numbers near the jet region in Case 2 is higher than that in Case 1. For Case 3, the Nusselt numbers near the jet region exhibit an increase compared with those in Case 0; this increase corresponds to the highest Nusselt numbers in Case 3. However, Nu numbers in the middle sub channel in Case 3 are less than those in Case 1 and Case 2. The endwall shape in Cases 4 and 5 is a broken line, and therefore, the Nu number distributions in Cases 4 and 5 are quite different from those in Cases 1, 2 and 3, and high Nusselt numbers are found near the junction of the broken line.

The laterally averaged Nu number in T6 are quantitatively described in Fig. 7. The highest Nu number can be observed near the jet region in Case 0. Then, the Nu numbers decrease gradually.

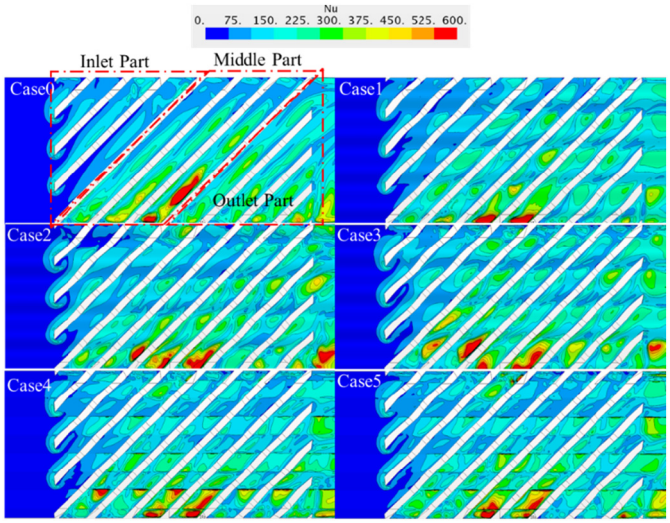


Fig. 6. Local Nu number distributions with different endwall shapes at $Re = 33{,}000$.

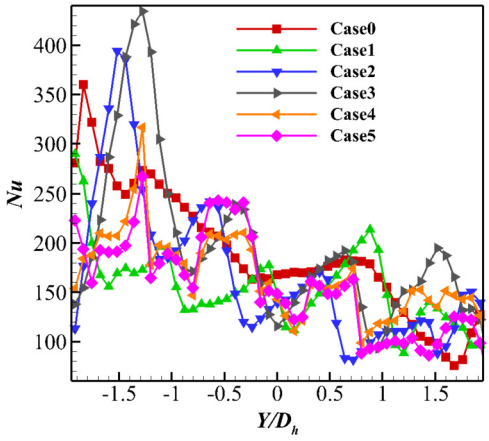


Fig. 7. Laterally averaged Nusselt numbers along the flow direction.

Nearby the turning region, the Nusselt numbers slightly increase; this trend is in accord with the experimental results [16]. Case 1 has lower Nusselt numbers near the jet region than Case 0; moreover, the Nu number decreases slowly in the flow direction. Thus, Case 1 shows a more uniform Nu number distribution than Case 0 does. For Case 2 and Case 3, the Nusselt numbers exhibit periodic fluctuations because of the endwall shape. In addition, Case 3 has a greater laterally averaged Nu number distribution than Case 2, which corresponds to the highest area-averaged Nu number in Case 3. For Cases 4 and 5, the Nusselt numbers also exhibit periodic fluctuations; however, the Nusselt numbers in Cases 4 and 5 are smaller than those in Cases 2 and 3.

4.3. Flow field performance

Fig. 8 shows the 3D distribution of streamlines in the latticework duct at $Re = 33{,}000$. For Case 0, the velocity increases sharply as the cooling air flows into the latticework. The high-velocity cooling air impinges into the sidewall and changes the flow direction toward the opposite sub channel near the turning region. Due to the crossing angle, the turning flow has a tangential velocity component. Under the effect of this tangential velocity and upward velocity component, the turning flow is converted into a spiral flow. Once the spiral flow arrives at the opposite channel, the cooling air changes the flow direction again. The upward velocity component of the spiral flow is transformed into axial velocity

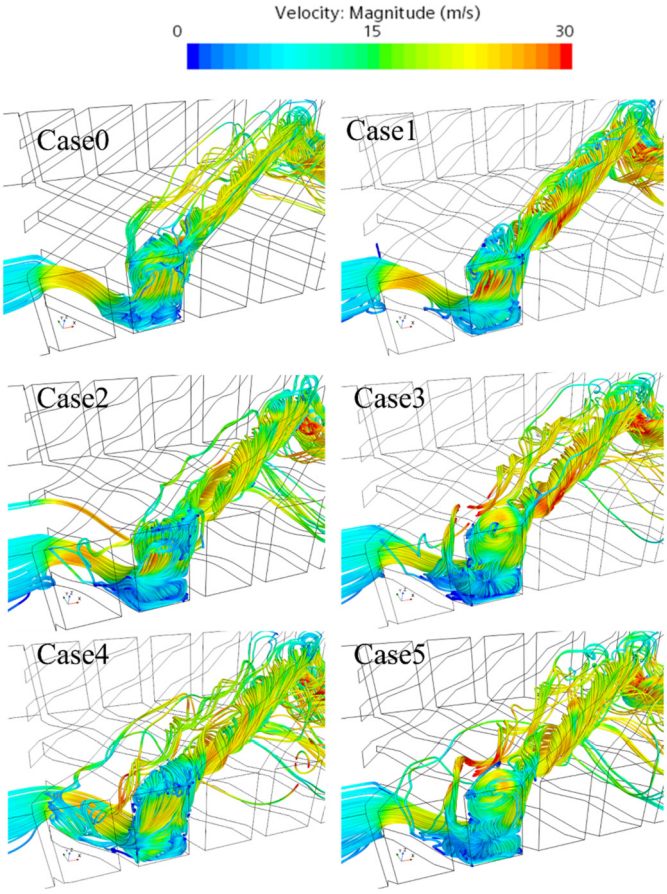


Fig. 8. A comparison of the 3D streamlines colored by the velocity at $Re = 33{,}000$.

component, whereas the tangential velocity component remains approximately constant. Therefore, a helical flow is formed. The above flow process is repeated in each sub channel. This repeated flow process contributes to heat transfer augmentation. In addition, some cooling air escape from the junction of the two layers of the sub channel. The escaped flow in the latticework duct, which is also referred to as the bypass flow, constitutes a very important flow structure [18]. The velocity in Z direction is also shown in Fig. 8. A negative value of the velocity is found at the junction between the LS and TS. The negative value means that some cooling air flows from the TS to the LS (bypass flow). In other cases, the helical flow, spiral flow and turning flow still dominate the flow fields. However, these flow structures display some different behaviors. For Case 1 and Case 2, the velocity increases near the sidewall; this increase in velocity weakens the bypass flow and enhances the jet effect. In addition, the axial velocity components of the helical flow in Case 1 and Case 2 are less than those in Case 0. For Case 3, upward spiral flow possesses a large tangential velocity component, and the helical flow is enhanced.

To illustrate the bypass flow and helical flow, Fig. 9 displays the velocity vector and value for the middle plane of T6. The streamlines in the figure show the helical flow, jet flow and turning flow. A large portion of the cooling air, which originates from L2 and L3, impinges onto T6. Then, the jet fluid changes direction and flows along the sub channel. However, some cooling air belonging to T6 is forced to the leading side by vortices (also called the bypass flow). These vortices are generated by the shear layer between the two sides of sub channel that correspond to the helical flow. Due to the bypass flow, the vortices become stronger. At T6, the cooling air also changes flow direction, as shown in Fig. 8. The flow structure exhibits some changes in response to the different rib shapes.

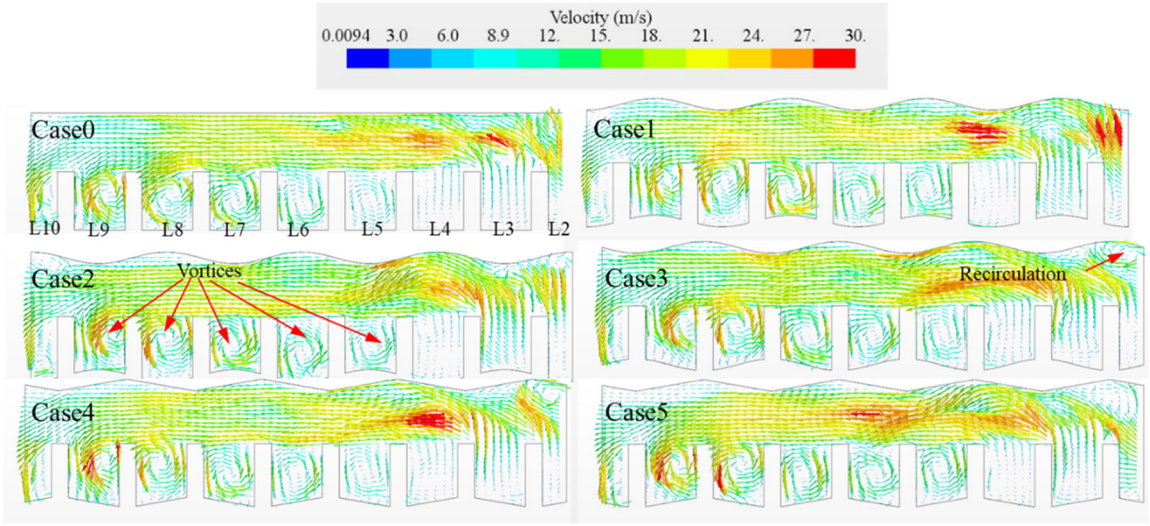


Fig. 9. A comparison of the velocity vector distributions on the middle plane of the T6 sub-channel.

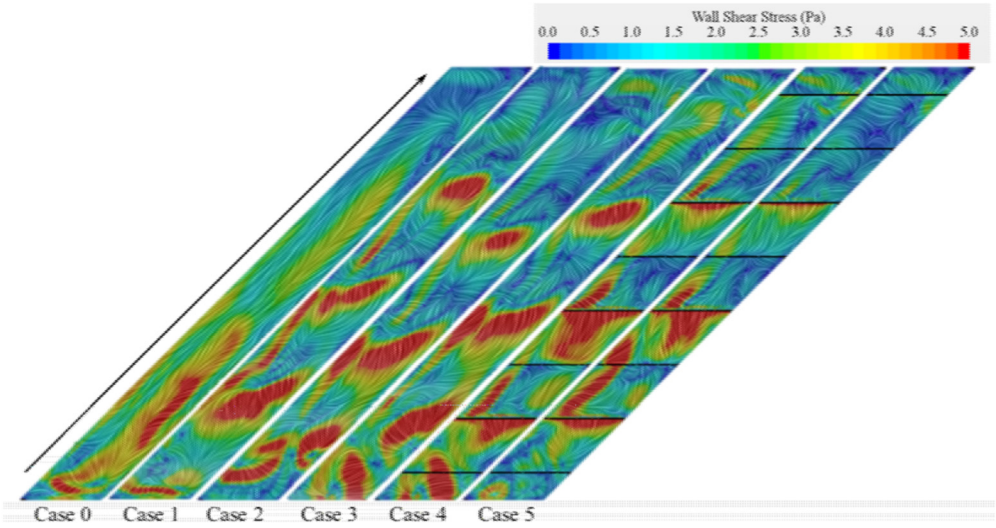


Fig. 10. Limiting streamline and τ distributions on the T6 sub-channel at $Re = 33{,}000$.

For Case 1, the flow in the beginning of the T6 sub channel has a high velocity, which corresponds to a strong jet effect. However, the flow in the turning region has a smaller velocity in Case 1 than in Case 0, which corresponds to a low Nusselt number, as shown in Fig. 7. The mainstream streamline in Case 2 obviously changes compared to those in Case 0 and Case 1. The helical flow acts as a cross-flow for a stronger jet flow of the L3 sub channel. The cross-flow in Case 3 is enhanced relative to that in Case 2 because of the high velocity above the L3 and L4. In addition, a large recirculation zone, which contains a low-energy fluid motion, is generated in the corner of the beginning of the T6 sub channel. A large recirculation is also found in Cases 4 and 5. However, Case 5 has the largest recirculation among the Cases 4 and.

Fig. 10 exhibits the limiting streamlines and wall shear stress (τ) distributions at $Re = 33{,}000$. The results show that the endwall shape has significant effects on the limiting streamline distributions. For Case 0, a reattachment line, which is caused by the jet, is found at the front of the T6. Between the sidewall and the reattachment line, a small recirculation is observed. In the mainstream, the reattachment line and separation are observed at corner of a sub channel, and a spiral point is found at the sub channel ending. The beginning of the sub channel has a relatively high wall shear stress, but the highest τ is found farther downstream of the reattachment line; thereafter, the τ decreases gradually. Therefore, the turning flow contributes to the small wall shear stress. For Case 1, the streamlines and τ vary significantly. The reattachment line is closer to the sidewall in Case 1 than in Case 0. The τ increases, especially near the bottom of the endwall. Within the turning region, the spiral point is farther away from the sidewall. For Case 2, the τ is obviously different at the jet region, and the high τ region is no longer visible. In addition, the τ increases near the turning region. For Cases 3 through 5, a high τ is found at the junction of the broken endwall. In addition, the τ decreases along the flow direction. The results further demonstrate that the endwall shape has little effect on the τ distribution in the middle part; in contrast, the τ near the jet region is closely related to the endwall shape. Case 4 has a greater τ near the jet region than Case 5. Within the turning region, the τ distributions are independent of the endwall shape.

Fig. 8 and Fig. 10 show that the endwall shape remarkably alters the flow structures near the turning and jet region. In order to explore the effects of the endwall shapes on the turning region and jet region in greater detail, Fig. 11 shows the streamlines and turbulent kinetic energy (TKE) distribution on the Y = 0.062 m plane. The results show that the streamlines and TKE distributions are closely related to the endwall shape. For Case 0, the helical flow

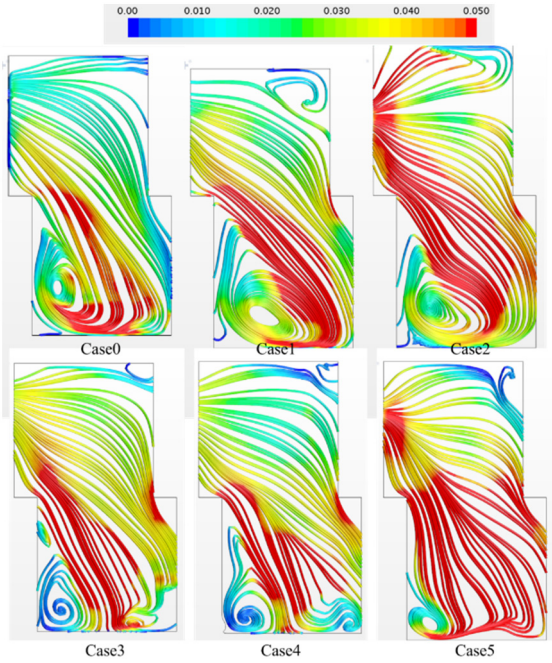


Fig. 11. Streamline and TKE distributions on the Y = 0.062 m plane at $Re = 33{,}000$.

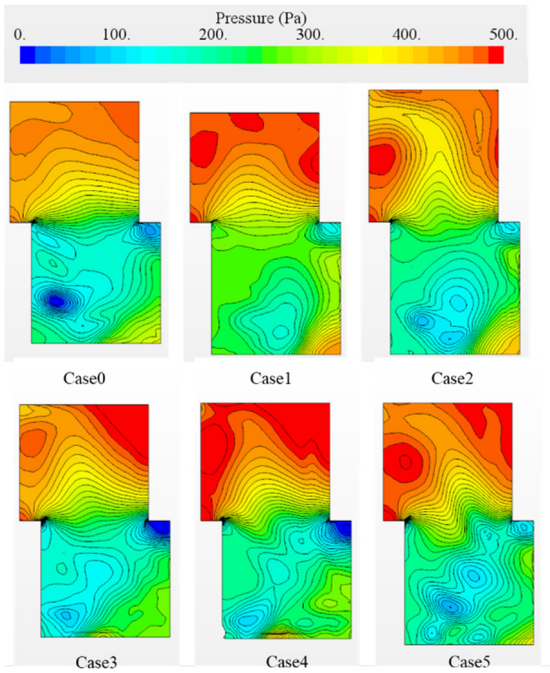


Fig. 12. Pressure distributions on the Y = 0.062 m plane at $Re = 33{,}000$.

impinges into the windward side and changes flow direction toward the leading side. Then, a spiral flow directly impinges toward the leading side. This strong jet effect induces a larger recirculation in the corner region of the leading side. Between the recirculation and jet flow, high TKE is generated because of the shear stress layer. The jet increases the heat transfer within the latticework duct. For Case 1, the small recirculation is generated on the trailing side because of the high adverse pressure gradient. In addition, the recirculation on the leading side becomes larger in Case 1 than in Case 0, which corresponds to a high TKE. For Case 2, the TKE obviously increases on the trailing side compared with the TKE distributions in Case 0 and Case 1. In addition, the recirculation on the leading side in Case 1 becomes smaller than that in Case 1. For Case 3 and Case 4, a small recirculation is observed on the leading side, corresponding to the recirculation shown in Fig. 9. This small recirculation weakens the jet impact. This weakened jet also induces a small recirculation in the left corner of the LS, which has already been shown in Case 0, Case 1 and Case 2. The TKE values in Case 3 and Case 4 also decrease compared with those in Case 1, Case 2 and Case 3. For Case 5, the recirculation decreases on the leading side, and the TKE grows abruptly.

Based on a previous study, the turning and jet regions cause high friction factor. Therefore, the pressure distribution on the Y = 0.062 m plane is shown in Fig. 12. The pressure sharply decreases at the junction between the two layers of the sub channel for all cases. For Case 0, the pressure distribution on the trailing side is relatively uniform. On the leading side, the lowest pressure is found near the core of the recirculation. The pressure slightly increases near the jet region. The reason for this increase in pressure is that a drop in kinetic energy leads to a rise of the static pressure near the jet region. The pressures on the trailing side in Case 1 and Case 2 are greater than that in Case 0; this increased pressure corresponds to a high friction factor. However, the low pressure near the recirculation is undetectable. This phenomenon means that the pressure on the leading side is more uniform in Case 1 and Case 2 than in Case 0. This finding also suggests that a suitable endwall shape could obtain a low friction factor. Due to the strengthened helical flow, the pressure at the windward side is increased for Case 3, Case 4 and Case 5. Furthermore, pressures near the jet region on the leading side in Cases 3, 4 and 5 are also less than those in Case 0, Case 1 and Case 2.

4.4. Thermal performances

Fig. 13 shows the distributions of $(Nu/Nu_0)/(f/f_0)$ and $(Nu/Nu_0)/(f/f_0)^{1/3}$ for various Reynolds numbers. Evidently, $(Nu/Nu_0)/(f/f_0)$ decreases when the Re number increases for all cases. This means that the increase of the friction factor is greater than the increase of the heat transfer ratio in the latticework duct. In addition, Case 5 provides the lowest $(Nu/Nu_0)/(f/f_0)$, while Case 3 provides the highest $(Nu/Nu_0)/(f/f_0)$. This finding indicates that changing the rib shape is an alternative method to increase $(Nu/Nu_0)/(f/f_0)$. Fig. 13(b) shows the $(Nu/Nu_0)/(f/f_0)^{1/3}$ distribution. Case 3 boasts the highest $(Nu/Nu_0)/(f/f_0)^{1/3}$, while Case 5 and Case 1 have the lowest $(Nu/Nu_0)/(f/f_0)^{1/3}$. Fig. 13(b) also proves that the rib shape could change the thermal performance significantly.

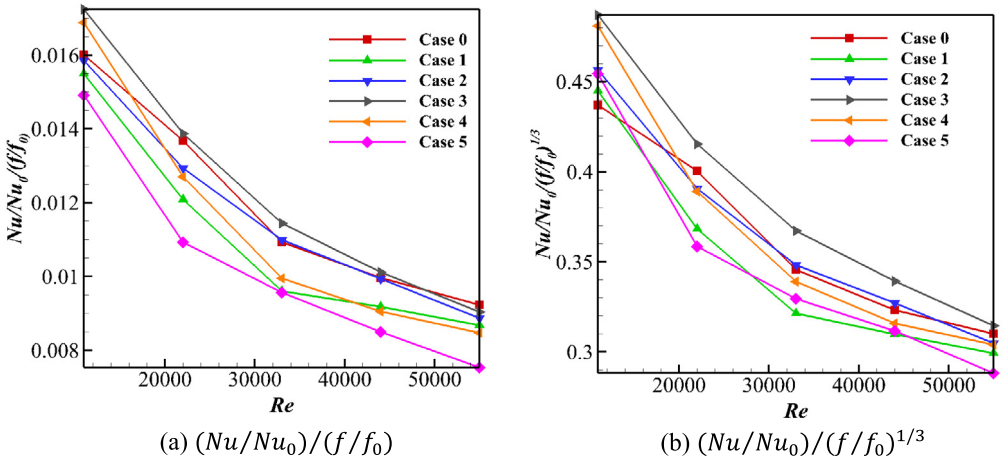
5. Conclusions

The effects of the endwall shape on the thermal performance in the latticework duct were studied by a commercial software. Different endwall shapes and different Re numbers were tested, and a polyhedral mesh was adopted. The main findings are summarized as follows:

(1) The helical flow, jet flow and turning flow were the most important flow features. A spiral flow could be observed near the sidewall. The jet flow and turning flow result in a high friction factor. In addition, the jet flow, which is near the beginning of the sub channel, resulted in the highest Nusselt number

(2) The jet flow and turning flow exhibited changes when different endwall shapes were adopted for the latticework duct. In addition, the endwall, which was shaped by the broken line, had a stronger cross-flow, which can weaken the jet effect.

(3) The friction factor was also affected by the endwall shape. In general, the traditional latticework had the lowest friction factor. After adopting different endwall shapes, the friction factor increased. The endwall shaped by the broken line had a higher



(a) $(Nu/Nu_0)/(f/f_0)$ (b) $(Nu/Nu_0)/(f/f_0)^{1/3}$

Fig. 13. Thermal performance in the latticework with various Reynolds numbers.

friction factor than the endwall shaped by the sine function and cosine function.

(4) The different endwall shapes also induced different Nu number distributions on the endwall surface. With the shaped endwall, the Nu number distribution exhibited periodic fluctuations. The results also showed that an appropriate endwall shape could simultaneously increase both the Nu number and the thermal performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests ar personal relationships that could have appeared to influence the work reported in this paper.

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